



HORIZON EUROPE
Research & Innovation Framework



GENERATIVE BIONICS
Physical AI • Robotics Intelligence

GENERATIVE BIONICS

Analisi Strategica del Fit con i Bandi Horizon Europe
2026-2027

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Cluster 4 – Digital, Industry and Space

Horizon Radar v1.1 • AHP + Gurobi LP

Executive Summary

FIT MEDIO CLUSTER

62.10%

CALL ANALIZZATE

1

Top 3 call

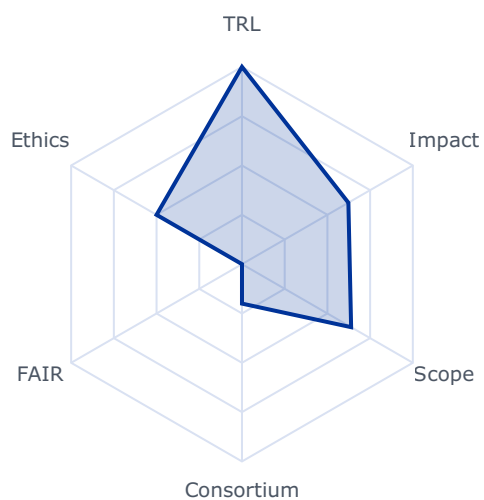
**#1 • HORIZON-CL4-2026-02-DIGITAL-
EMERGING-53-two-stage**

GO

Innovative AI methods and technologies for the process
industries (RIA) (Processes4Planet ...

62.1 / 100

Radar complessivo cluster



Distribuzione Top 10

#1 62.1

Rank #1 • HORIZON-CL4-2026-02-DIGITAL-EMERGING-53-two-stage

GO

Innovative AI methods and technologies for the process industries (RIA) (Processes4Planet and AI, Data and Robotics partnerships)

Fit score: 62.10/100 • Solver: Optimal

Perché questa call ha preso 62/100

La call HORIZON-CL4-2026-02-DIGITAL-EMERGING-53-two-stage ottiene un fit del 62.1% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 30.6/100). Solver: Optimal, CR AHP: 0.0168.

*"Projects are expected to contribute to the following outcomes:"**"To develop and demonstrate innovative AI-driven solutions in materials and process"**"development to enhance the competitiveness and sustainability of the process industries"***Gap principali da colmare**

- ⚠ Vincolo gender balance attivo sul piano di implementazione

Azioni concrete (prossimi 3-6 mesi)

- ✅ Formalizzare policy e target gender balance su team/proposal

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

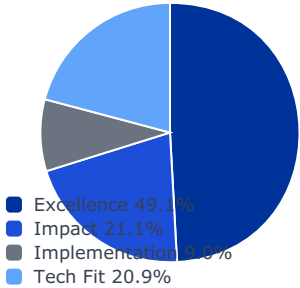
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

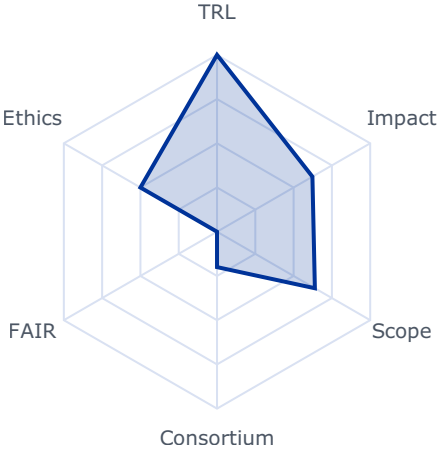
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6228	0.6228	0.3060
impact	0.2105	0.6933	0.6933	0.1459
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6388	0.6388	0.1333

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 0.0
- budget_max: 8000000.0
- sme_required: False
- sme_ok: False
- ssh_required: False
- gender_balance_required: True

Expected Outcomes

Projects are expected to contribute to the following outcomes:

- To develop and demonstrate innovative AI-driven solutions in materials and process development to enhance the competitiveness and sustainability of the process industries by better products and process technologies and reducing the time to market
- To increase the competitiveness of materials production in Europe by AI-supported optimal operation of plants and value networks and early detection of problems and failures
- To improve the working conditions in the plants by using AI technologies, metaverse, and robots.

137 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

Scope

Drastically improved AI methods and technologies hold transformative potential for the process industries, enabling advancements in process design, operational efficiency, and faster innovation across the entire lifecycle of plants and products. By using different AI approaches such as multimodal generative AI, foundation models, and agentic AI, the industry can move beyond conventional AI applications as e.g. predictive maintenance and quality control toward more intelligent, adaptive, and creative solutions.

AI-based solutions can deliver value at every stage of the process lifecycle. In design and engineering, they can enable new innovations and accelerate the development process. During operations, AI technologies can be employed to optimize processes, enhance reproducibility, adapt to changing conditions, and provide new forms of support for the workforce and enable autonomous operation. In value chains, AI can help to adapt faster and detect changing customer needs. These capabilities support faster innovation and strengthen competitiveness in a rapidly evolving industrial landscape. However, realizing this potential requires careful consideration of risks related to reliability, security, and trust, ensuring that AI solutions are effective, safe, and responsible.

Proposals should produce dedicated innovative AI-based solutions for the process industry for one of the following scopes:

- More effective and faster development of new materials and processes
- Competitive and sustainable production, reducing the negative environmental impact of industry
- Reduction of risks for the health of the workforce and for the environment and making workplaces in the process industries more attractive.

In the projects, user acceptance and training of the users as well as integration into the OT/IT landscape of the companies should be taken into account, e.g. through active engagement in design, development and integration of systems and processes.

This topic is linked to the Apply AI Strategy, therefore proposals should seek collaboration with relevant initiatives.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to Destination 'Leadership in materials and production for Europe'.

This topic implements the co-programmed European Partnerships Processes4Planet and AI, Data and Robotics.

Disclaimer

Report generato automaticamente da Horizon Radar v1.0 – Strumento decisionale basato su Analytic Hierarchy Process (AHP) + Linear Programming (Gurobi). I risultati sono indicativi e supportano la prioritizzazione strategica.

Generative Bionics R&D Team



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